

CONTROLLED BENCHMARK

Qwen Local vs GLM 5.3

A frozen-evidence comparison of grounded QA and structured Function Documentation in CodeCompass

Publication package for thesis evaluation, academic appendix, and supplementary technical reporting.

Benchmark	Repositories	Search	QA	Documentation
controlled_benchmark_v1	3	54 executions	6 cases	6 cases

Compared systems

Local baseline: Ollama with qwen2.5-coder-3b-codecompass:latest. External comparison: OpenAI-compatible glm-5.3-flash. The only intended changing variable was the generation provider/model.

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1. Executive Summary

This controlled experiment evaluates whether changing only the language-model provider/model changes CodeCompass answer quality. Repository snapshots, indexes, embeddings, retrieval evidence, prompts, contexts, citations, generation parameters, and human evaluation criteria were fixed.

Grounded QA: GLM returned five reviewable responses with 10.0/10 correctness and groundedness. Its sixth QA output was truncated and remains INCONCLUSIVE. On those same five pairs, Qwen averaged 7.6/10 correctness and 8.8/10 groundedness.

Function Documentation cannot support a numeric quality comparison. Qwen produced six scored outputs with an average of 6.5/10. All six GLM outputs were INCONCLUSIVE: four empty-content provider responses and two truncated structured outputs.

The results show stronger GLM grounded-QA generation on the five reviewable pairs, alongside a serious GLM Documentation reliability limitation. They do not establish universal superiority or statistical significance.

2. Experimental Setup

Repository dataset

Repository	Commit	Files	Symbols	Chunks	Vectors	Valid
Hospital-System	da0b1f8cf04f36d6281a5fd44b797ad195147633	33	409	409	409	Yes
CS-Bookstore	56ee148f3b4b8bc1f4eaa921e22f280fa14ad7a	56	144	144	144	Yes
CodeCompass	9e6ed02bf5edc4eb14341bfa838e4bfd10c92f1d	96	1220	1220	1220	Yes

Every frozen index had exact SQLite/Chroma ID agreement, matching generation markers, and vector_complete=true.

Benchmark size and configuration

- 54 frozen search executions: 18 bilingual cases across lexical, semantic, and hybrid retrieval.
- 6 grounded-QA cases and 6 Function Documentation cases.
- Embedding: Ollama nomic-embed-text-local:latest, 768 dimensions.
- QA retrieval: hybrid, limit 5, maximum context 6,000 characters.
- Generation: temperature 0, maximum 1,200 tokens, zero provider retries and zero manual reruns.
- Token usage: unavailable for both provider paths.

3. Methodology

Retrieval was executed once and persisted. Each model received the same stored system prompt, user prompt, context, evidence, and trusted citation metadata. Prompt and evidence hashes support integrity verification.

QA review measured correctness, groundedness, citation accuracy, completeness, and hallucination. Documentation review measured purpose, parameters, return value, behavior, dependencies, citation accuracy, unsupported claims, and overall quality.

Failed, empty, malformed, or truncated outputs remain INCONCLUSIVE. They are reported as reliability outcomes and are not assigned estimated semantic scores.

The design isolates generation changes, but provider and model changed together. It therefore cannot attribute a failure uniquely to model internals versus provider behavior.

4. Search Evaluation

Search is shared and model-independent in this experiment because retrieval evidence was frozen before generation.

Method	Target present	Top-10 rate	Interpretation
Lexical	13/18	72.2%	Shared frozen retrieval
Semantic	8/18	44.4%	Shared frozen retrieval
Hybrid	14/18	77.8%	Shared frozen retrieval

Source navigation resolved for 527/527 returned records. Manual relevance scores were not measured, so expected-target presence is reported instead of a general retrieval-accuracy score. No model-dependent search change can be claimed.

5. QA Evaluation Comparison

Per-case scores are correctness / groundedness / citation accuracy. The Persian CodeCompass GLM case is retained as INCONCLUSIVE because finish_reason=length produced a truncated answer.

Case	Repo	Lang	Qwen	GLM	Q comp.	G comp.	Q hall.	G hall.
CB-QA-H-EN	Hospital-System	EN	8/10/10	10/10/10	partially complete	complete	none	none
CB-QA-H-FA	Hospital-System	FA	6/7/10	10/10/10	partially complete	complete	major	none
CB-QA-B-EN	CS-Bookstore	EN	6/8/10	10/10/10	partially complete	complete	minor	none
CB-QA-B-FA	CS-Bookstore	FA	9/9/10	10/10/10	complete	complete	none	none
CB-QA-C-EN	CodeCompass	EN	9/10/10	10/10/10	partially complete	complete	none	none
CB-QA-C-FA	CodeCompass	FA	4/6/10	INCONCLUSIVE	partially complete	incomplete	major	none

Paired reviewable cases

Metric	Qwen (n=5)	GLM (n=5)
Correctness	7.6/10	10.0/10
Groundedness	8.8/10	10.0/10
Citation accuracy	10.0/10	10.0/10
Complete answers	1/5	5/5
Hallucination findings	2/5	0/5

Across all six Qwen QA cases, correctness averaged 7.0/10, groundedness 8.3/10, and citation accuracy 10.0/10. Qwen had one minor and two major hallucination findings. GLM had five PASS and one INCONCLUSIVE; its 10.0 averages apply only to reviewable cases.

6. Documentation Evaluation Comparison

Case	Repo	Lang	Qwen	GLM	Qwen finding	GLM finding
CB-DOC-H-EN	Hospital-System	EN	7	INCONCLUSIVE	unsupported side effect	no reviewable output
CB-DOC-H-FA	Hospital-System	FA	9	INCONCLUSIVE	None recorded	structured output truncated
CB-DOC-B-EN	CS-Bookstore	EN	9	INCONCLUSIVE	None recorded	no reviewable output
CB-DOC-B-FA	CS-Bookstore	FA	7	INCONCLUSIVE	details omitted	structured output truncated
CB-DOC-C-EN	CodeCompass	EN	7	INCONCLUSIVE	details omitted	no reviewable output
CB-DOC-C-FA	CodeCompass	FA	0	INCONCLUSIVE	truncated invalid JSON	no reviewable output

Qwen's measured overall Documentation score was 6.5/10 across six cases. Its citation score averaged 10.0/10. Weaknesses included an unsupported side effect, missing implementation details, and one truncated invalid JSON result.

GLM Documentation quality is unavailable. Four cases returned invalid_response_empty_content; two returned incomplete JSON with finish_reason=length. No purpose, parameter, return, behavior, dependency, citation, or overall score was estimated.

7. Latency and Runtime Analysis

Model	Attempts	Mean	Median	Min	Max	Tokens
Qwen local	12	46.271s	45.810s	9.167s	94.510s	Unavailable
GLM 5.3	12	8.078s	9.818s	2.795s	12.462s	Unavailable

GLM attempts were faster, but seven of its twelve outputs were not reviewable. The latency difference is therefore an attempt-latency observation, not successful-task throughput. Qwen executed locally with no cloud-generation cost; GLM cost was not captured.

8. Error Analysis

Qwen generation findings

- Hospital English QA omitted appointments.
- Hospital Persian QA described a literal empty-queue result as an error.
- Bookstore English QA omitted the superclass save call and invented uniqueness.
- CodeCompass English QA omitted unique/sorted behavior.
- CodeCompass Persian QA contained a wrong or garbled service-flow explanation.
- Documentation included one unsupported state update, missing details, and one truncated invalid JSON output.

GLM execution findings

- One QA case ended with finish_reason=length and remained INCONCLUSIVE.
- Four Documentation cases returned invalid_response_empty_content.
- Two Documentation cases ended with finish_reason=length and incomplete JSON.
- No failed Documentation output received an estimated semantic score.

Expected targets were present in 35/54 frozen search executions and in 14/18 hybrid executions. This means both retrieval and generation limitations are visible. Different answers from identical contexts support generation as a contributor where evidence was sufficient, but aggregate results do not justify assigning every failure to generation.

9. Scientific Findings

Changing the provider/model improved grounded-QA quality on the five reviewable pairs: GLM was more complete and received higher correctness and groundedness scores. The result is bounded to this frozen sample.

GLM was not universally more suitable. Its single QA truncation and complete lack of reviewable Documentation outputs expose provider/output-format reliability limitations. Qwen was slower and less accurate in QA, but remained operationally usable for five of six Documentation cases and offered offline, private execution without cloud generation cost.

Trusted citation validity remained intact where CodeCompass metadata was available, even when generated content was incomplete or wrong. This demonstrates separation between deterministic citation integrity and semantic answer quality.

Limitations

- Small sample: six QA and six Documentation cases across three Python repositories.
- No statistical significance analysis.
- One embedding model and one retrieval configuration.
- Provider and model changed together.
- Token and cost measurements unavailable.
- No reviewable GLM Documentation cases.
- Results must not be generalized beyond the tested workloads.

10. Reproducibility Appendix

Frozen artifact hashes

Artifact	SHA-256
benchmark_cases.json	5fbec49e1ad4c70af6a8aabb028f473afbdeb807d189070914b778ed3e9699af
frozen_retrieval_evidence.json	2359b07c36f19d47faf0171de0ab5e48ebc8b2f4620b6a8a8a6865cf75cc4c83
qwen_results.json	f0ed5404b5a7b86f33147ac002841f42624c173807325abfc32faa62f00d4f86
qwen_quality_evaluation.json	e048e00fa3f1c38aebd261a9914fc1bce2987eba0d8b83a82c1fda7c8fddeccd
glm_results.json	a2802c6f598c36baf9496cf915491d977b8abb1f39811de7f7e8f2435c43aaa2
glm_quality_evaluation.json	9966990408e7cd503ef34fc0078ec786fea386caee538dcef17dc5b66cf727bf

Repository pins

Repository	Commit
Hospital-System	da0b1f8cf04f36d6281a5fd44b797ad195147633
CS-Bookstore	56ee148f3b4b8bc1f4eaa921e22f280fa14ad7a
CodeCompass	9e6ed02bf5edc4eb14341bfa838e4bfd10c92f1d

Artifact locations and dates

- Canonical benchmark: reports/evaluation/controlled_benchmark_v1/benchmark_cases.json
- Frozen evidence: reports/evaluation/controlled_benchmark_v1/frozen_retrieval_evidence.json
- Raw model results: qwen_results.json and glm_results.json
- Human evaluations: qwen_quality_evaluation.json and glm_quality_evaluation.json
- Execution dates: unavailable because the authoritative JSON artifacts do not record them.
- Report generation date: 2026-09-02.

No indexing, retrieval, generation, rescoring, prompt change, or source-code change was performed to create this report.